

10. (a) Global warming is believed to be mainly the result of increasing levels of carbon dioxide in the atmosphere.

- (i) State how the balance of carbon dioxide and oxygen is maintained in the atmosphere. Explain why levels of carbon dioxide are increasing and how this leads to global warming. [3]

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- (ii) Carbon capture and storage (CCS) is one possible method of reducing global warming.

Describe briefly how CCS is carried out. [2]

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(b) Oxides of nitrogen are released in vehicle exhaust fumes. They can cause smog in busy city centres as well as acid rain.

(i) One oxide of nitrogen contains 30.4 g of nitrogen and 69.6 g of oxygen.

Find the simplest formula of this oxide. You **must** show your working. [3]

$$A_r(\text{N}) = 14$$

$$A_r(\text{O}) = 16$$

Simplest formula

(ii) The oxide with this simplest formula is found to have a relative molecular mass of 92. Find its molecular formula. [2]

Molecular formula

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